

FRONTEND ENGINEERING

The DOM & Event Propagation

Understanding interaction mechanics in the browser

The DOM Structure

Building the logical tree of a web document

What is the DOM?

The Document Object Model (DOM) is a programming interface for web documents. It represents the page so that programs can change the document structure, style, and content.

 DOM tree structure diagram

- ❑ Tree Structure: Every element is a node in a parent-child hierarchy.
- ❑ Live Interface: Changes in the DOM are instantly reflected in the browser.
- ❑ Language Neutral: While usually JS, it can be accessed by other languages.

Accessing the DOM



ID Selector

Fastest access using
`getElementById()` . Returns a
single unique element.



Query Selector

Modern access using CSS-
style selectors with
`querySelector()` or
`querySelectorAll()` .



Class & Tag

Select multiple elements
using
`getElementsByClassName()` or
`getElementsByTagName()` .

DOM Manipulation

Changing Content

Use `textContent` to update plain text or `innerHTML` to inject HTML structures. Always prefer `textContent` for security unless tags are required.

Creating Elements

Dynamically generate nodes with `createElement()` and insert them into the tree using `appendChild()` or `prepend()`.

Modifying Styles

- ❌ Inline Styles: Modify specific CSS properties directly using the `element.style.property` syntax.
- ❌ Class Management: Best practice is to use `classList.add()`, `remove()`, or `toggle()` to switch pre-defined CSS classes.
- 🎯 Computed Styles: Retrieve actual rendered styles using `getComputedStyle(element)` for layout calculations.

Event Mechanics

Handling user interaction and event flow

Event Listeners

The Listener

Use `addEventListener()` to attach a function to an event. This method allows multiple listeners for the same event on a single element.

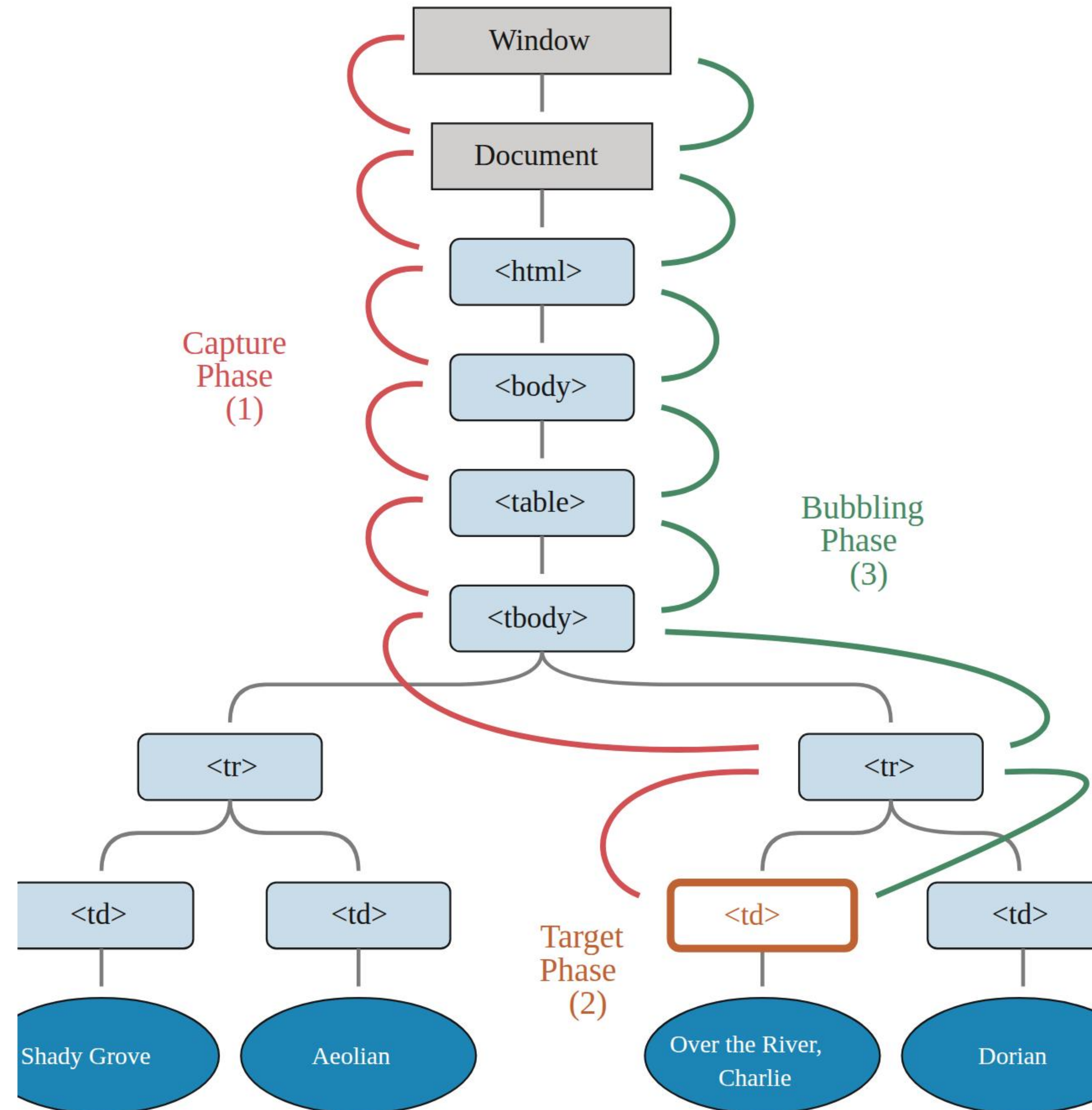
Common Events

Respond to actions like `click`, `submit`, `keydown`, and `mouseover`. Each event object contains details like target and mouse position.

Event Propagation

Propagation is the journey an event takes through the DOM tree. It consists of three phases: Capture, Target, and Bubble.

Bubbling is the default behavior where an event starts at the target element and moves up to the root, triggering parent listeners along the way.



Propagation Flow

Phase	Direction	Description
Capturing	Downwards	Event travels from Window down to the target element.
Target	At Source	The event reaches the actual element that triggered it.
Bubbling	Upwards	Event travels from target back up through parents to Window.

Control Propagation

Stopping the Bubble

Call `event.stopPropagation()` to prevent the event from triggering listeners on parent elements. Essential for nested interactive components.

Default Behavior

Use `event.preventDefault()` to stop the browser's standard response, such as page reloads on form submissions or link navigation.

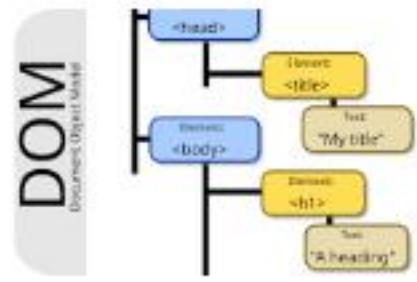
Questions?

Mastering DOM structure and event flows



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Image Sources



<https://upload.wikimedia.org/wikipedia/commons/thumb/5/5a/DOM-model.svg/1280px-DOM-model.svg.png>

Source: en.wikipedia.org



<https://javascript.info/article/bubbling-and-capturing/eventflow.svg>

Source: javascript.info